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Network Traffic Monitoring and Analysis Tool Using Python

Innovative Evaluation Practices for BCSE308P – Computer Networks Lab

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Problem and Motivation

- Every network interaction is carried by packets across the TCP/IP layers.
- Raw dumps (tcpdump) are accurate but overwhelming — no aggregated view.
- Capture needs elevated privileges and a live network just to look.
- Goal: readable insight and explainable alerts, demonstrable offline.

Existing Approach (Baseline) and Its Gap

- tcpdump / libpcap: accurate capture, CLI only, no analysis or detection.
- Wireshark: deep but heavyweight; Scapy: a library, not a dashboard.
- Signature IDS (Snort/Suricata): opaque to learners, heavy to deploy.
- Gap: a lightweight, transparent, offline-capable teaching analyser.

Proposed Novelty

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- One normalised, header-only PacketRecord feeds all nine dashboard views.
- Five explainable, configurable rules — reason and severity per alert.
- Fully offline demo mode with injected ground-truth anomalies.
- Refined SYN per-pair port-scan rule removes false positives.

Architecture / Algorithm

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- source → threaded CaptureWorker → bounded thread-safe queue.
- parse_packet → PacketRecord (memory-bounded) → pandas DataFrame.
- analyzer + anomaly detector + filters + report generator.
- Presented through a nine-page Streamlit dashboard.

Implementation Setup

- Python 3.11 with Scapy, Streamlit, pandas, Plotly and SQLite.
- Eleven modules with clean separation of capture, analysis and UI.
- 37 pytest tests using synthetic packet fixtures (no privileges).
- Reproducible demo generator with a fixed random seed.

Scenarios and Metrics

- Load scenarios: S1 = 693, S2 = 2,293, S3 = 8,293 packets.
- Throughput (packets/second) and processing latency (milliseconds).
- Detection coverage and false positives on clean traffic.
- Deterministic, repeated runs reported as mean ± SD.

Results (Baseline vs Proposed)

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- Throughput rises from ~29k to ~76k pkt/s as load scales.
- Injected anomalies detected: 4 of 4 (3 of 4 at high load).
- Port-scan false positives: 3 / 8 / 8 → 0 / 0 / 0 after refinement.
- Analysis comfortably exceeds typical classroom capture rates.

Analysis and Limitations

- pandas has fixed overhead — raw counting is faster but gives no analysis.
- Spike rule is relative to mean, so fixed bursts are less visible at high load.
- Header-only: encrypted HTTPS/TLS payloads are never inspected.
- Switched-LAN visibility and NAT-based direction inference are limited.

Conclusion, Future Work and References

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- A working, tested and honest traffic monitoring and analysis tool.
- Zero false positives on clean traffic at every tested load scale.
- Future: GeoIP, TLS SNI parsing, ML-based detection, live BPF filters.
- References: Tanenbaum; Kurose & Ross; Scapy; Streamlit; pandas.